

Design Plan

Jeremiah Dados, Dmytro Stavskyi, and Luis Wong

1. Introduction

a. Purpose / Need

The project is designed for use by researchers, specifically those in the Neural Interface Technology lab at UF, to facilitate the conduct of animal experiments involving Differential Tissue Coupled Powering (DTCP) powered implants more efficiently. To power and interface with this type of implant, there are several pieces of lab equipment necessary, which are discrete (as in separate) elements that also need to be interfaced with each other. With this design, the functionalities provided by these tools would be implemented within a single compact PCB device, which would both simplify the overall setup for conducting experiments and provide reliability as the PCB would be a newly manufactured device. Furthermore, it will have additional features that provide functionality beyond the simple powering of the implant.

b. Domain & Prior Art

The project is intended for use in the field of neuroscience, in particular for the use with the interfacing and the powering of implants. The areas of focus for improvement include the capabilities of the implants themselves, such as their power consumption and size, but also external interfacing with the implants. Some particular examples of existing designs similar to ours or that address a similar issue include PCBs for powering brain implants [1] and a transmitter designed to be embedded with an implantable medical device [2]. The differentiating features of our design include the EMG recording system, which will record the electrical activity of the muscles from the RF pulses for feedback and possible storage, and the power amplifier used to increase the transmission output power from 1 W to 4 W. In addition, our system is separate from the implant, and so it does not require intrusive means to be attached to the animal body in question or be embedded with the implant.

c. Impact & Risk Assessment

The development of the proposed PCB-based DTCP platform aims to extend the capabilities demonstrated in [3] by providing a compact, safe, and fully integrated system for in vivo and in vitro research, much akin to how Analog Discovery, made by Digilent, works to compact the many devices used by electrical and computer engineers. As is apparent from the description we have of this device thus far, the device intends to impact the scientific community. The intended scientific impact includes streamlined validation of tissue-coupled powering efficiency, improved quantification of implant performance, and reduced experimental variability. By integrating multiple functions into a single compact PCB, the project reduces the need for several discrete and costly lab instruments, lowering the overall financial burden on research facilities. Furthermore,

concerning environmental impact, the system reduces the material footprint associated with manufacturing and maintaining multiple separate pieces of lab equipment. The project’s broader, societal impact lies in supporting the development of fully injectable, wireless electroceutical devices that can enable minimally invasive neural interfacing for therapeutic applications [3].

Potential risks associated with this device primarily relate to electromagnetic safety, hardware reliability, and ethical compliance. DTCP operates in the MHz range, where tissue heating and unintended stimulation can occur if current density exceeds safe limits [3, Sec. II-A]. Additionally, electromagnetic interference (EMI) may couple into nearby circuits or distort bioelectric recordings if insufficient shielding or grounding is not implemented. Hardware faults could result in unstable power output, which may damage the implanted device or produce misleading data. Ethical risks include compliance with the Institutional Animal Care and Use Committee (IACUC) standards for all animal procedures [3, Sec. II-E]. By prioritizing adherence to IACUC standards and non-invasive power delivery, we reinforce the cultural expectations surrounding the humane treatment of research animals.

2. Statement of work

a. Performance specifications

The device that will be built for this project will meet the following specifications.

Specification	Proof-of-concept value	Target value
RF output frequency	10 MHz ~ 30 MHz	10 MHz ~ 30 MHz
RF output power	1 W	2 W
Idle power consumption	100 mW	10mW
Tx peak power consumption	4 W	8 W
Battery life	10 min.	24 hr.
Board dimensions	70 mm x 50 mm	40 mm x 30 mm

Table 1: Hardware Specifications

Note that there are two categories of requirements: proof-of-concept for a “minimum” version of the device, and a target for the best performance and usability. Because this is an innovative research project, it might be impossible to meet the target values within two semesters - in that case, meeting the proof-of-concept goals might still be enough to produce the desired research output.

Specification	Value
Software platform	Texas Instruments MSP ARM Cortex M0 microcontroller
Language	Embedded C
Build environment	Texas Instruments Code Composer Studio

Table 2: Software Specifications

b. Core features, target completion dates, and initial assignment of tasks

The high-level project plan for Fall 2025 and Spring 2026 is presented in the task table below:

Tasks	Finish by	Leader	Supporters	Comments
First Prototype				The first prototype aims to realize the most basic functionality: 25 MHz differential output at 1W.
Design the PA circuit	End of October 2025	Luis Wong	Jeremiah Dados	
Design the EMG circuit	End of October 2025	Dmytro Stavskyi	Jeremiah Dados	
Design the first prototype PCB in Altium	Mid- November 2025	Jeremiah Dados	Luis Wong, Dmytro Stavskyi	
Place the fabrication order	Mid- November 2025	Jeremiah Dados	Han Wu	
Receive the board and assemble it	End of November 2025	Jeremiah Dados	Luis Wong, Dmytro Stavskyi	
Validate board outputs on lab equipment	Mid- December 2025	Jeremiah Dados	Luis Wong, Dmytro Stavskyi	Present the functional prototype at the end of semester.
Validate board functionality in animal experiments	End of January 2026	Han Wu	Jeremiah Dados, Luis Wong, Dmytro Stavskyi	Work resumes the second week of January after the winter break.
Second Prototype				The second prototype addresses the issues found in the first prototype and makes the device more compact and flexible.
Revisions to the circuit design	Mid-February 2026	Luis Wong, Dmytro Stavskyi	Jeremiah Dados	
Design and order the PCB	End of February 2026	Jeremiah Dados	Luis Wong, Dmytro Stavskyi	
Place the fabrication order	End of February 2026	Jeremiah Dados	Han Wu	
Receive the board and assemble it	Mid-March 2026	Luis Wong, Dmytro Stavskyi	Jeremiah Dados	
Validate board outputs on lab equipment	End of March 2026	Jeremiah Dados	Luis Wong, Dmytro Stavskyi	
Validate board functionality in animal experiments	End of April 2026	Han Wu	Jeremiah Dados, Luis Wong, Dmytro Stavskyi	

Table 3: Project plan with approximate task deadlines

It is expected that two iterations of prototyping can be completed between October 2025 and April 2026 due to the biomedical nature of the work and possible delays caused by how challenging RF design is. Naturally, the pace of iteration and work will highly depend on the results of testing the first prototype.

3. Deliverable artifacts

The following artifacts will be delivered at the end of the project

a. The final PCB

The final device will be a fully assembled and characterized board. Several instances of the PCB will be prepared for use at the Neural Interface Technology Lab. The PCB will be physically located at the Lab and will remain useful for many years without requiring maintenance. The presence of multiple boards will allow researchers to continue the experiments even if some devices become damaged.

b. The device firmware

The firmware (a Code Composer Studio Project) for the microcontroller used on the board will be maintained through a GitHub repository. The repository is also how the code will be distributed for Neural Interface Technology Lab researchers. The firmware does not depend on changing APIs or databases, so the last version of it will remain usable for many years. The code will be written in a way that makes it easy to fit into future applications.

c. Documentation

The documentation containing information on how to use the device will be provided in the firmware repository and distributed alongside it.

4. Mockups

a. Interfaces

The external systems the device will interact with are the following:

- The firmware will be developed on PCs running the Code Composer Studio IDE. The IDE allows the user to directly program a microcontroller on a custom board as long as the PC running the IDE is connected to a Launchpad through USB, and the Launchpad is connected to the microcontroller through a 3-wire programming interface.

- A TI LaunchPad board will be connected to the DTCP board through a 3-wire TI programming interface (SWCLK, SWDIO, RST) using jumper wires and an on-board connector. This will allow the Code Composer Studio IDE to recognize the microcontroller on the board as a programmable device and upload the firmware to it.
- The board will be powered by a 3.7V Li-ion battery, connected to the VCC and GND terminals of the device.
- The board connector will also have debugging pins to test the voltages on key traces of the PCB. Laboratory equipment like RF oscilloscopes will be connected to these pins to test the design.

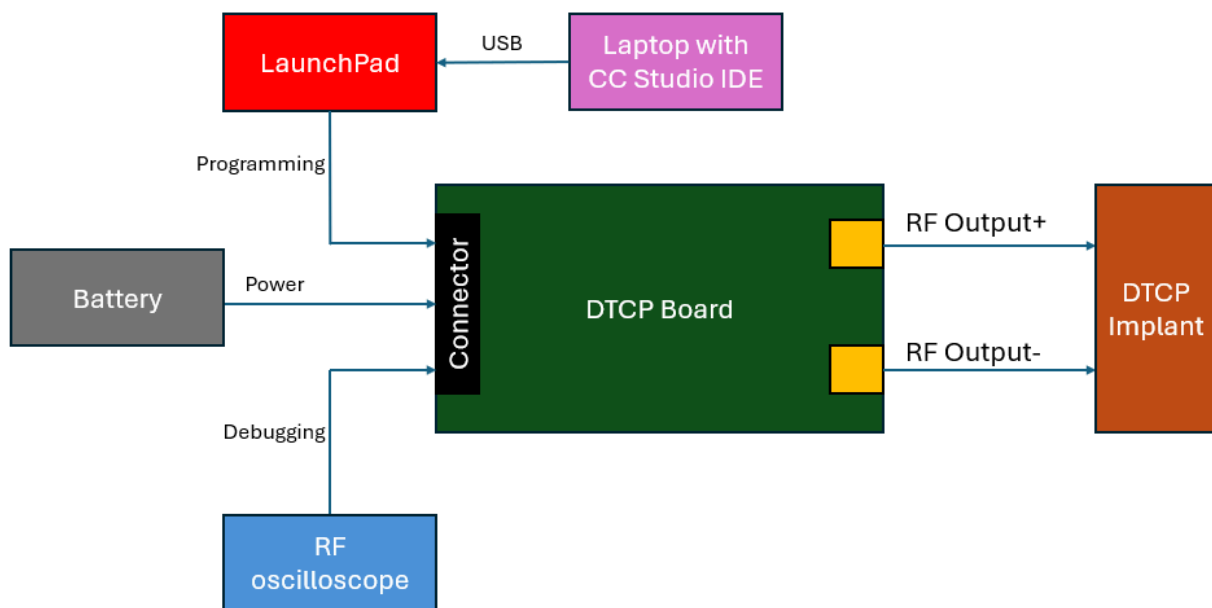


Figure 1: A high-level diagram of the interfaces in the project

b. Systems

The board will consist of the following sub-systems:

- The power circuit converts the 3.7V from the battery into 3.3V for the MCU and the transmitter IC and 12V for the PA.
- The ARM Cortex M0 MCU programmed by the Launchpad / Code Composer.
- An RF transmitter IC, likely a DDS (Direct Digital Synthesizer) or a Radio module controlled by the microcontroller.
- RF power amplifier increases the power of the signal from ~10 dBm to over 30 dBm.
- A balun RF circuit turning the output into a differential signal.
- An EMG AFE (analog front-end) chip measures muscle activity in response to the stimulation and relays the data to the microcontroller.

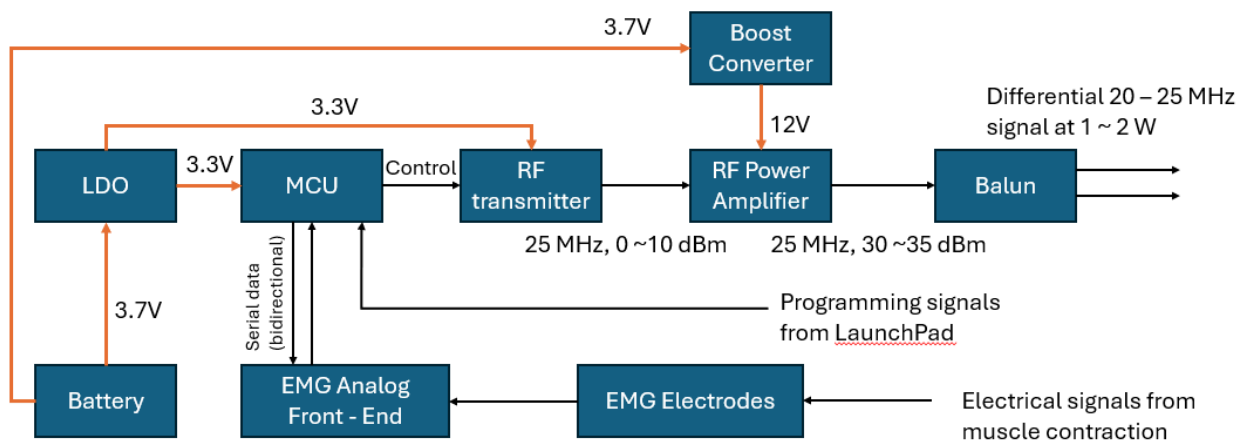


Figure 2: Hardware Sub-Systems

The specific parts or detailed circuits will be chosen later on in the design process, guided by research and testing.

c. Networking

The only communications involved in the project will be within the proprietary TI Launchpad programming interface.

d. Storyboards

N/A: There will be no interactive/visual front-end.

e. Draft Schematics

The following in-progress schematics show early versions of the PA and EMG subsystems:

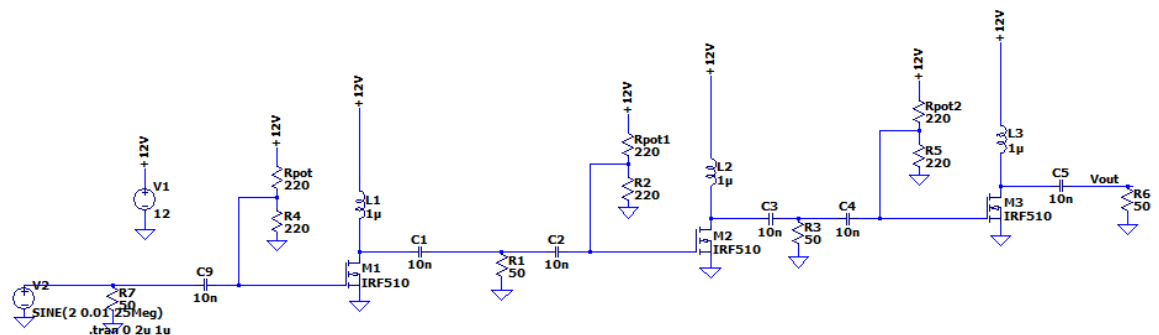


Figure 3: Early version of a three-stage class-A RF amplifier circuit to be potentially used in the project

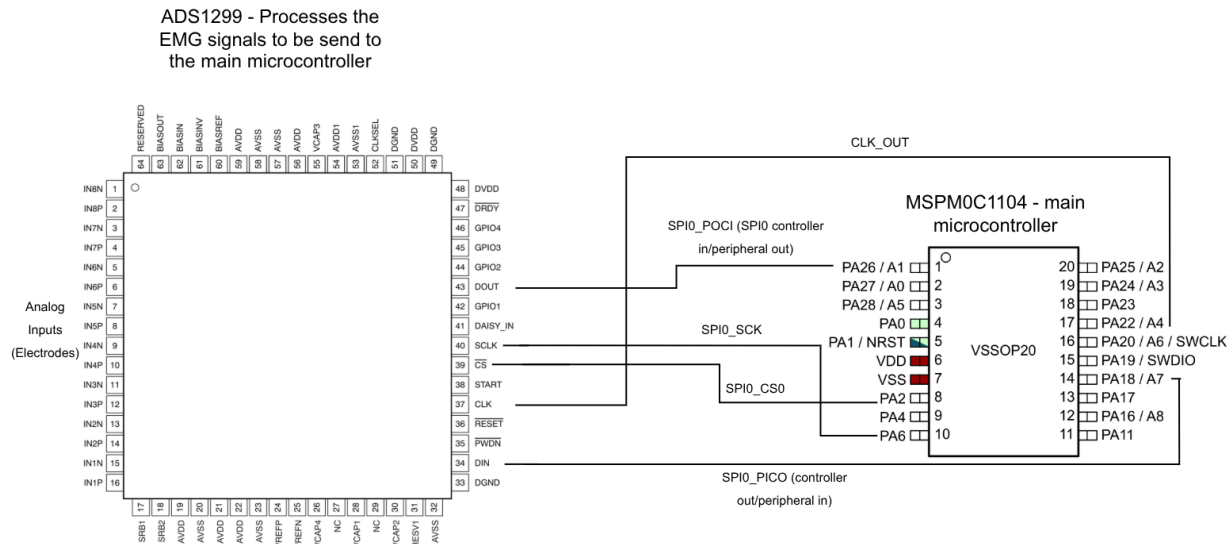


Figure 4: Schematic of the EMG circuit connected to the microcontroller

The above schematics represent preliminary research and the final versions (including all other sub-systems) will be available for the pre-alpha build.

References

- [1] J. Dados, R. Faherty, G. Summers, H. Wu, K. Burstiner, and A. Khalifa, "A Compact Sub-Scalp Device for Powering Brain Implants," 2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), Orlando, FL, USA, 2024, pp. 1-4, doi: 10.1109/EMBC53108.2024.10782386.
- [2] F. Goodarzy and E. Skafidas, "A fully integrated 200 μ W, 40pJ/b wireless transmitter for implanted medical devices and neural prostheses," 2013 35th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), Osaka, Japan, 2013, pp. 3246-3249, doi: 10.1109/EMBC.2013.6610233.
- [3] H. Wu, A. I. Omi, S. Mahmud, A. Jiang, M. Halac, C. Wallace, E. Patrick, S. Srinivasan, B. Chatterjee, and A. Khalifa, "Differential Tissue Coupled Powering for Injectable Electroceuticals," unpublished manuscript, University of Florida and Harvard University, 2025.